EQUINOX

Conversion Rate Optimization — A/B Test Plan

Mobile Checkout Completion for First-Time Shoppers (Ages 25–40)
Prepared by: CRO Lead | Test Window: 4 Weeks | Channel: Mobile Web/App Checkout

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1. Test Objective

Primary Test Objective

Increase the completed purchase rate among first-time mobile visitors during the checkout flow, specifically by reducing abandonment at the shipping and payment steps, within the four-week test window.

Target Audience Definition

- **Segment:** First-time visitors (no prior purchase or account history) on Equinox's DTC site
- **Age range:** 25–40 years old
- **Device:** Mobile web and in-app checkout sessions only (desktop traffic excluded from this test)
- **Product context:** Premium fitness apparel purchases, typical cart value consistent with full-price and light-promo SKUs

Funnel Problem Statement

Analytics indicate a disproportionate drop-off between cart initiation and order confirmation on mobile, concentrated at two points: (1) the shipping information step, where address entry and delivery method selection introduce friction, and (2) the payment step, where checkout form length, perceived security, and payment method rigidity appear to drive hesitation and exit. First-time visitors are the most affected segment because they lack saved payment credentials, account history, or established trust with the brand, making friction and trust signals more decisive at the point of purchase.

2. Hypotheses

Primary Hypothesis

If Equinox reduces friction in the mobile checkout flow at the shipping and payment steps, **then** completed purchase rate among first-time mobile visitors will increase, **because** removing unnecessary form fields and decision points lowers the effort and hesitation that currently causes abandonment before payment confirmation.

Supporting Hypotheses

- **H2 — Form simplification:** If the shipping form is reduced to essential fields with autofill/address lookup enabled, then shipping-step abandonment will decrease, because fewer manual inputs reduce mobile typing friction and errors.

- **H3 — Trust signaling:** If recognizable trust cues (secure checkout badge, return policy reminder, customer review snippet) are made more visible near the payment button, then payment-step completion will improve, because first-time buyers lack prior brand trust and rely on visible reassurance at the moment of payment.
- **H4 — Payment flexibility:** If additional express payment options (e.g., digital wallet one-tap checkout) are surfaced above manual card entry, then overall mobile purchase completion will increase, because express options reduce the number of steps between intent and confirmed purchase.

Hypothesis-to-Metric Mapping

Hypothesis	Primary Metric Impacted	Funnel Step Measured
H1 — Primary (overall friction reduction)	Mobile purchase completion rate	Cart → Order confirmation
H2 — Form simplification	Shipping-step completion rate	Shipping information step
H3 — Trust signaling	Payment-step completion rate	Payment step
H4 — Payment flexibility	Checkout-to-purchase conversion rate	Payment step → Confirmation

3. Variants

Experiment Scope Statement

This test isolates a single experiment theme — **checkout friction reduction** — to ensure results can be attributed cleanly to the tested change. No unrelated layout, pricing, promotional, or navigation changes will be introduced during the test window.

Control vs. Variant Definition

	Control (A)	Variant (B)
Shipping form	Full form: name, address line 1/2, city, state, zip, phone, delivery notes (8 fields)	Condensed form with address autocomplete/autofill; delivery notes fiel
Payment step	Standard card-entry form; trust badges below the fold	Trust badge, secure-checkout icon, and free-returns note repositioned
Payment methods	Manual card entry only, digital wallet available but not surfaced as the primary option	Digital wallet one-tap option (Apple Pay/Google Pay) surfaced as the p
Steps in flow	3 steps: Shipping → Payment → Review	3 steps: Shipping → Payment → Review (same structure, reduced inp

Note: *All three changes (form simplification, trust signaling, payment flexibility) roll up under the single "reduced checkout friction" theme and will be tested together as one combined Variant B against the current Control, so that the four-week window yields one clear, actionable read rather than three underpowered sub-tests.*

4. Success Metrics

Primary Metric

Mobile purchase completion rate — completed orders divided by checkout-initiated sessions, among first-time mobile visitors.

Supporting Metrics

- Shipping-step completion rate (progression from shipping form to payment step)
- Payment-step completion rate (progression from payment step to order confirmation)
- Cart abandonment rate (sessions that reach cart but never initiate checkout)
- Time-to-checkout-completion (median session duration from cart to confirmation)

Guardrail Metrics

- Average order value (AOV) — must not decline meaningfully as a side effect of the simplified flow
- Return/refund rate — must remain stable, ensuring simplified shipping input does not increase address errors
- Payment failure/error rate — must not increase with the new express payment option
- Customer support contact rate related to checkout — must not increase post-change

5. Test Rules

Rule	Specification
Traffic allocation	50% Control / 50% Variant, randomized at the session level for first-time mobile visitors only; assignment pers
Test duration	Full 4-week window, no early stopping based on interim peeks; minimum run time ensures at least 2 full week
Minimum sample size	Sized to detect a relative lift of 8–10% in mobile purchase completion rate at 95% confidence and 80% power,
Statistical significance threshold	95% confidence level ($p < 0.05$) on the primary metric before declaring a winner
Decision rule	Declare Variant B the winner only if the primary metric shows a statistically significant improvement AND all gu
QA requirement	Full cross-device QA (iOS Safari, Android Chrome, in-app WebView) completed before launch; no mid-test ch

6. Analysis Plan

Result Analysis Framework

Compare Control vs. Variant B on the primary metric (mobile purchase completion rate) using a two-proportion significance test. Supporting metrics are reviewed directionally to explain and validate the primary result (e.g., confirming the lift originates at the shipping and payment steps as hypothesized, not from unrelated traffic shifts). Guardrail metrics are reviewed as pass/fail checks rather than optimization targets.

Segmentation Checklist

- First-time mobile shoppers overall (primary read)
- Device type: iOS vs. Android, and mobile web vs. in-app, to check for platform-specific effects
- Traffic source: paid social, paid search, organic/direct, to rule out acquisition-mix skew
- Age sub-bands within 25–40 (e.g., 25–32 vs. 33–40) to check for consistency across the target segment
- New-visitor vs. returning-but-non-purchaser sessions, to confirm the effect is concentrated in true first-time buyers

Action Criteria

- **Ship Variant B:** Primary metric shows statistically significant positive lift, guardrails stable → roll out to 100% of mobile first-time-visitor traffic.
- **Inconclusive:** Primary metric is directionally positive but not statistically significant → extend test or re-run with a larger/longer sample rather than shipping or discarding.
- **Do not ship:** Primary metric is flat or negative, or any guardrail metric shows significant degradation (e.g., AOV drop, higher payment failure rate) → revert to Control and revisit the hypothesis, isolating individual elements (form, trust cues, payment method) in a follow-up test to identify the underperforming piece.

This plan is intended for execution handoff to the Growth and Analytics teams. All test parameters (allocation, duration, and decision thresholds) should be locked prior to launch and not modified mid-test.